Talha Khan

734-740-1486 EE.TalhaKhan@gmail.com linkedin.com/in/talhakhan-ee talhaakhan2006.github.io

Summary

Electrical Engineering student at the University of Michigan–Dearborn with experience leading multidisciplinary engineering teams developing autonomous vehicle systems. Skilled in embedded systems, electrical integration, and wire harness design while managing a $30,000 engineering project.

Education

University of Michigan–Dearborn **Expected May 2027**
Bachelor of Science in Engineering in Electrical Engineering | GPA: 3.87 *Dearborn, MI*

Wayne County Community College **May 2024**
Associate of Science | GPA: 3.95 *Belleville, MI*

Experience

Intelligent Systems Club — Project Phoenix **Aug 2025 – Present**
Chief Engineer & Treasurer *University of Michigan–Dearborn*

- Lead the electrical, mechanical, and software integration of an autonomous go-kart that earned **3rd Place** at the Autonomous Karting Series (AKS) National Grand Prix.
- Manage a $30,000 project budget, overseeing procurement, planning, and resource allocation while keeping the project under budget and on schedule.
- Direct multidisciplinary engineering efforts across a 20-member student team through design reviews, integration, and project planning.

Teaching Assistant (Grader) **Jan 2025 – Present**
ECE 270, ECE 273, ECE 3731 *University of Michigan–Dearborn*

- Independently managed feedback to students by addressing questions, clarifying course material, and providing constructive guidance to improve understanding.
- Review and grade assignments, quizzes, and examinations for ECE courses while providing technical feedback on circuits, embedded systems, and programming.

FIRST Robotics Team 6190 **Sept 2021 – May 2026**
Mentor

- Mentored a 20+ member FIRST Robotics team in CAD, electrical wiring, fabrication, and system integration.
- Taught industrial safety standards, electrical wiring, and pneumatic systems to high school team members with zero prior engineering experience.

Projects

Project Phoenix – Autonomous Go-Kart | *Electrical Integration* **2024 – Present**

- Designed and integrated the vehicle's electrical wire harness.
- Developed electrical routing, termination, and power distribution.
- Integrated embedded sensors and motor controllers during system validation.

Autonomous Mobile Robot | *Arduino Framework* **Fall 2024**

- Designed a line-following autonomous mobile robot with obstacle avoidance capabilities.
- Focused on achieving high speed and stability while carrying a payload.

Adaptive Light System | *ECE 210* **Fall 2025**

- Designed an automatic lighting system using a CdS photoresistor and op-amp comparator.
- Reduced estimated energy consumption by 30–35%.

Technical Skills

Programming: Python, C, JavaScript, HTML/CSS
Engineering: Embedded Systems, Wire Harness Design, Electrical Integration, Circuit Design, Power Distribution, Autonomous Systems, PCB Design
Software: Fusion 360, Altium Designer, LTspice, AutoCAD, Linux, Git
Hardware: Oscilloscopes, Function Generators, Multimeters, Soldering, Arduino, STM32, HCS12

Awards & Activities

Dean's List | Emerging Scholar | Mentor Collective Certification (Diversity, Equity & Inclusion) | Tau Beta Pi Engineering Honor Society | IEEE | Intelligent Systems Club | Phi Theta Kappa | Muslim Student Association